

NICHOLAS BORRELLO

AI & Software Engineer — Agentic Systems & LLM Applications

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SKILLS & CAPABILITIES

AI & AGENTS LLM agents • Model Context Protocol (MCP) • Tool calling • Structured outputs • Agent orchestration • Model routing/fallback • RAG • LLM evaluation • VLM/OCR	LANGUAGES & ML TypeScript • Python • Java • SQL • PyTorch • CNNs • Transformers • spaCy • MediaPipe • Computer vision • Document intelligence	FULL STACK & INFRASTRUCTURE React • Next.js • Node.js • Bun • Hono • PostgreSQL • SQLite • Supabase • Docker • Playwright • GitHub Actions • Azure DevOps
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EXPERIENCE

Bay State Pet & Garden — AI & Software Engineer ◆ Develop an AI Product Intelligence system that researches external product data, resolves exact product/variant identity, and produces evidence-backed catalog changes for human review. ◆ Engineer the agent execution layer with 25 bounded research tools, model routing and fallback, privacy controls, tool/model budgets, deadlines, and fail-closed validation. ◆ Maintain the retail CMS and AI evaluation pipeline, testing new extraction and agent behavior against verified datasets before staged rollout into product onboarding and publishing workflows.	Taunton, MA (Remote) Jul 2024 – Present
Allegro MicroSystems — Process Engineering Intern ◆ Developed automated test and deployment services using Python, Java, and Docker to standardize driver updates across engineering equipment. ◆ Maintained Azure DevOps CI/CD pipelines and Node.js automation utilities, replacing manual deployment processes.	Manchester, NH Jun 2023 – Aug 2023

EDUCATION

Worcester Polytechnic Institute — M.S. Computer Science	Worcester, MA Aug 2025
Worcester Polytechnic Institute — B.S. Computer Science, <i>With Distinction</i>; Dean's List (Fall 2022)	Worcester, MA May 2024

SELECTED PROJECTS

Resumancer AI — Agentic Career Workspace ◆ Built a career workspace comparing job requirements with a Master CV, proposing evidence-backed changes with reviewable diffs instead of unconstrained rewrites. ◆ Added ATS compatibility checks, model fallback, and PDF export so verified career data moves from job analysis to an application-ready resume.	Oct 2025 – Present Next.js, TypeScript, Vercel AI SDK, Supabase
ShopSite MCP Server — E-Commerce Tool Integration ◆ Built an MCP server that lets AI agents retrieve orders, search products, and manage inventory in ShopSite without interacting directly with legacy APIs. ◆ Wrapped XML/CGI requests, HMAC-SHA1 authentication, and legacy response formats behind typed, validated tools designed for safe LLM use.	Jan 2026 TypeScript, Model Context Protocol, Zod, Node.js
American Sign Language Gesture Recognition ◆ Trained CNN & Transformer models to recognize 100+ ASL gestures using ResNet34 and MediaPipe landmark sequences. ◆ Engineered temporal sequence extraction pipelines and parallelized data preprocessing using Python multiprocessing.	Jan 2025 – May 2025 Python, PyTorch, ResNet34, MediaPipe
NEVI Search Tool — Policy Document Intelligence ◆ Built an NLP-assisted document search tool to help analysts locate policy evidence across lengthy state NEVI plans. ◆ Implemented exact keyword search, fuzzy matching, configurable term groups, and spaCy preprocessing to speed policy-document review.	Aug 2022 – Dec 2022 Python, spaCy, PyMuPDF, PyQt6